10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

# Chapter 3 — Human Blood Circulatory System

Solved Exercise | New Book (PECTAA)

## Exercise (Solved)

### A. Select the correct answers for the following questions.

**1. Which component of blood is mainly responsible for carrying oxygen?**

(a) White blood cells (b) Platelets
(c) Red blood cells (d) Plasma

**Answer: (c)** — Red blood cells contain haemoglobin, which combines with oxygen and becomes bright red. Only about 1.5 % of the oxygen is carried dissolved in plasma.

**2. A person with low platelet count is likely to face difficulty in:**

(a) Fighting infections (b) Breathing
(c) Clotting blood (d) Transporting oxygen

**Answer: (c)** — Platelets gather at a damaged blood vessel and convert the plasma protein fibrinogen into fibrin, which forms a net that traps red blood cells and makes a clot. If the platelets are few, the blood does not clot and bleeding continues. Fighting infection is the work of white blood cells and carrying oxygen is the work of red blood cells.

**3. Which of the following layer is found in all blood vessels?**

(a) Smooth muscle (b) Endothelium
(c) Skeletal muscle (d) Connective tissue

**Answer: (b)** — The inner layer of endothelium is present in arteries, veins and also in capillaries, whose wall is made of only this single layer of cells. Smooth muscle and connective tissue are absent in capillaries.

**4. Which of the following contains deoxygenated blood in an adult human?**

(a) Left atrium (b) Pulmonary artery
(c) Pulmonary vein (d) Aorta

**Answer: (b)** — In an adult all arteries carry oxygenated blood except the pulmonary artery, which carries deoxygenated blood from the right ventricle to the lungs. Similarly, all veins carry deoxygenated blood except the pulmonary veins.

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**5. When fibrinogen makes blood clot it separates from blood and the rest is:**

(a) Plasma (b) Lymph
(c) Serum (d) Pus

**Answer: (c)** — When fibrinogen is removed from the blood plasma during clotting, the remaining liquid is called serum. So serum is plasma without the clotting proteins.

**6. Which heart chamber has the thickest walls?**

(a) Right atrium (b) Left atrium
(c) Left ventricle (d) Right ventricle

**Answer: (c)** — The left ventricle is the largest, thickest and strongest chamber of the heart, because it has to pump the oxygenated blood into the aorta and from there it has to reach all parts of the body against a high pressure.

**7. The exchange of materials between the blood and tissues occurs in:**

(a) Arteries (b) Veins
(c) Capillaries (d) Arteries and veins

**Answer: (c)** — The wall of a capillary is only a single layer of endothelium, which is so thin that water, nutrients and oxygen pass through it into the tissue fluid, and carbon dioxide and other wastes pass from the tissue fluid into the blood.

**8. All veins carry deoxygenated blood except the _____ vein.**

(a) Vena cava (b) Hepatic portal
(c) Pulmonary (d) Renal

**Answer: (c)** — The pulmonary veins bring oxygenated blood from the lungs to the left atrium of the heart, so they are the exception. The vena cavae, the hepatic portal vein and the renal veins all carry deoxygenated blood.

**9. Myocardial infarction is due to blockage of blood flow in:**

(a) Aorta (b) Pulmonary artery
(c) Coronary artery (d) Hepatic artery

**Answer: (c)** — The coronary arteries supply blood to the heart muscles. When one of them is blocked, usually by a blood clot over an atherosclerotic plaque, the heart muscle gets no blood and part of it dies. This death of heart muscle is called myocardial infarction or heart attack.

**10. If the bicuspid valve is damaged, which flow of blood will be affected?**

(a) From left atrium to left ventricle (b) From right atrium to right ventricle
(c) From left ventricle to aorta (d) From lungs to left atrium

**Answer: (a)** — The bicuspid (mitral) valve is present between the left atrium and the left ventricle. If it is damaged, it does not close properly when the ventricle contracts, so blood flows back into the left atrium and the flow from the left atrium to the left ventricle is affected.

## B. Write short answers.

### 1. List the types of cells present in blood along with their functions.

| Cell / cell-like body | Function |
|---|---|
| Red blood cells (erythrocytes) | Contain haemoglobin, which transports oxygen from the lungs to all body cells and also carries a little carbon dioxide. |
| White blood cells (leukocytes) | Defend the body against diseases. Neutrophils destroy bacteria, eosinophils fight parasites, basophils cause inflammation and allergic reactions, monocytes make macrophages which engulf germs, and lymphocytes make antibodies. |
| Platelets (thrombocytes) | Convert fibrinogen into fibrin at a damaged vessel, so that a clot is formed and bleeding stops. |

Blood also contains plasma, which is not a cell but the liquid portion. It carries nutrients, wastes, hormones and salts, and about 55 % of the blood is plasma.

### 2. Differentiate between atria and ventricles.

| Atria | Ventricles |
|---|---|
| They are the upper chambers of the heart. | They are the lower chambers of the heart. |
| Their walls are thin, because they only push blood into the ventricles. | Their walls are thick and muscular, because they pump blood to the lungs and to the whole body. |
| The right atrium receives deoxygenated blood from the body through the vena cavae, and the left atrium receives oxygenated blood from the lungs through the pulmonary veins. | The right ventricle pumps deoxygenated blood into the pulmonary arteries, and the left ventricle pumps oxygenated blood into the aorta. |
| Blood enters them during diastole, when they are relaxed. | They contract during systole and pump blood out of the heart. |

### 3. Trace the flow of blood from the vena cava to various body parts.

The superior and the inferior vena cavae bring deoxygenated blood from the body into the right atrium. The right atrium contracts and sends this blood into the right ventricle, which pumps it into the pulmonary arteries. These arteries take the blood to the lungs, where carbon dioxide diffuses out of the blood and oxygen diffuses into it.

The pulmonary veins now carry this oxygenated blood back to the left atrium of the heart. From the left atrium the blood goes into the left ventricle, and when the left ventricle contracts it pumps the blood into the aorta. The aorta gives branches which carry the blood to the head, shoulders, arms and the lower parts of the body,

and from the smallest arteries the blood reaches the tissues through the capillaries.

Vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary vein → left atrium → left ventricle → aorta → body parts

### 4. Identify the causes and treatments of myocardial infarction.

**Cause:** myocardial infarction means the death of heart muscles, and it is commonly called a heart attack. It occurs when the blood flow to a part of the heart muscle is blocked for a prolonged period. This blockage is usually caused by a blood clot that forms over the atherosclerotic plaques inside a coronary artery, so that part of the heart muscle gets no oxygen and dies.

**Symptoms:** chest pain with tightness and pressure, shortness of breath, sweating, and pain that radiates to the arm, neck or jaw.

**Treatment:** emergency treatment with medicines that dissolve the clot, angioplasty to open the blocked artery, and stenting to keep the artery open. The patient is also advised to keep the weight normal, avoid smoking, exercise regularly and control blood pressure, cholesterol and diabetes, because otherwise it can lead to heart failure, arrhythmias and sudden cardiac arrest.

### 5. Colour the diagram of heart to show oxygenated and deoxygenated blood and label the structures.

In the diagram, the parts that contain deoxygenated blood (dark red / blue) are the superior vena cava, the inferior vena cava, the right atrium, the right ventricle and the pulmonary artery. The parts that contain oxygenated blood (bright red) are the pulmonary veins, the left atrium, the left ventricle and the aorta.

| Structure | Blood carried |
|---|---|
| Superior and inferior vena cavae, right atrium, right ventricle, pulmonary arteries | Deoxygenated blood |
| Pulmonary veins, left atrium, left ventricle, aorta | Oxygenated blood |

The valves are also labelled: the tricuspid valve between the right atrium and the right ventricle, the bicuspid valve between the left atrium and the left ventricle, the semilunar valves between the ventricles and the large vessels, and the septum between the left and the right sides of the heart.

## C. Write answers in detail.

### 1. Describe the functions of the components of blood.

Blood is a connective tissue which transports materials through the body. About 55 % of it is plasma and 45 % is made of cells or cell-like bodies. An adult human has about 5 litres of blood.

**Plasma:** it is the liquid portion of the blood and is composed of 90–92 % water, 7–9 % proteins and about 1 % other substances. Antibodies (plasma proteins) defend the body against pathogens, fibrinogen is responsible for blood clotting and albumin maintains the osmotic pressure of blood. The plasma salts are mainly sodium, chloride and bicarbonate ions, and plasma also carries nutrients (glucose, lipids,

amino acids) from the digestive system, wastes from the cells, hormones from the endocrine glands, and small amounts of carbon dioxide and oxygen.

**Red blood cells:** they are biconcave, disc-shaped cells which contain haemoglobin. They transport oxygen from the lungs to all the body cells and carry a little carbon dioxide. There are 4–5.5 million RBCs per $mm^3$ of blood and their average life span is about 120 days, after which they are removed by the spleen and the liver.

**White blood cells:** they are colourless, irregular and fewer in number (about 7000 per $mm^3$). They defend the body against diseases: neutrophils destroy bacteria, eosinophils fight parasites, basophils release histamine and cause allergic reactions, monocytes make macrophages which engulf germs and dead cells, and lymphocytes make antibodies.

**Platelets:** they are small, nucleus-less fragments of the large cells of the bone marrow—there are about 250,000 per $mm^3$ and their life span is 7–12 days. When a vessel is damaged they gather at the site and convert fibrinogen into fibrin, whose net traps RBCs and forms a clot that stops the bleeding.

### 2. Explain how blood transports materials throughout the human body.

Blood is the transport system of the body, and it works together with the heart and the blood vessels. The heart works as a double pump: its right side collects blood from the body and sends it to the lungs, while its left side collects blood from the lungs and sends it to the body.

**Transport of oxygen and carbon dioxide:** blood takes oxygen from the lungs and carries it to all the body cells, where the oxygen is used in cellular respiration. The carbon dioxide produced by the cells is carried back by the blood to the lungs, from where it is removed in the exhaled air.

**Transport of nutrients:** digested food such as glucose, amino acids, glycerol and fatty acids is absorbed in the small intestine into the blood. From the intestine this blood goes to the liver through the hepatic portal vein, and from the liver to the heart and then to all parts of the body.

**Transport of hormones and wastes:** hormones secreted by the endocrine glands are carried by the blood to their target organs. Wastes such as urea and carbon dioxide are carried from the cells to the kidneys and the lungs for excretion.

**Other functions:** blood distributes heat in the body, its white blood cells and antibodies defend the body against pathogens, and its platelets stop bleeding by forming clots. In this way blood keeps every part of the body supplied with what it needs and removes what it does not need.

### 3. Explain the structure of the heart with a diagram.

The heart is a muscular organ which lies in the chest cavity, beneath the sternum and between the two lungs. A tough sac-like membrane called the pericardium surrounds it and secretes a fluid which reduces friction between the pericardium and the heart. The wall between the left and the right sides of the heart is called the septum, and each side is divided into two chambers. The upper chambers are the atria and the lower chambers are the ventricles; the atria have thinner walls than the ventricles.

**Valves:** special flaps which open in only one direction are present between the chambers. The valve between the right atrium and the right ventricle is the tricuspid valve (three flaps) and the valve between the left atrium and the left ventricle is the bicuspid valve (two flaps). When the ventricles contract, these valves close so that blood cannot flow back into the atria. A semilunar valve is present between each ventricle and the large vessel leaving it, and it prevents the blood from flowing back into the ventricle.

**Working:** the right atrium receives deoxygenated blood from the body through the superior and inferior vena cavae; the right ventricle pumps it into the pulmonary arteries, which take it to the lungs. The oxygenated blood from the lungs returns through the pulmonary veins into the left atrium, goes to the left ventricle, and the left ventricle pumps it into the aorta, which distributes it to the whole body. In this way the heart works as a double pump, and the flow of blood from the heart to the lungs and back is called pulmonary circulation, while the flow from the heart to the body tissues and back is called systemic circulation.

### 4. Explain the harmful effects of smoking related to heart diseases.

Smoking is the major risk factor for the development of heart diseases. Its harmful effects on heart health are:

**(i)** Smoking accelerates atherosclerosis by damaging the endothelium of the arteries, so fatty plaques are deposited inside them and the arteries become narrow and hard.

**(ii)** It increases the chances of the blood clotting by promoting platelet aggregation, and these clots can block the coronary arteries and cause myocardial infarction (heart attack).

**(iii)** It causes vasoconstriction, that is, narrowing of the blood vessels, which raises the blood pressure.

**(iv)** It increases the heart rate and disturbs the heart rhythm, causing arrhythmias (irregular heartbeats) which can be dangerous.

**(v)** Carbon monoxide present in the smoke binds to haemoglobin more strongly than oxygen does, so it reduces the oxygen-carrying capacity of the blood and less oxygen reaches the heart muscle.

**(vi)** Smoking also narrows the arteries of the legs, which puts extra strain on the heart. Non-smokers who inhale the smoke of others have 30 % more chances of heart disease, so passive smoking is also dangerous.

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### 5. Compare the structure and function of an artery, a vein and a capillary.

| Feature | Artery | Vein | Capillary |
|---|---|---|---|
| Function | Carries blood away from the heart | Carries blood towards the heart | Exchanges materials between blood and tissues |
| Wall | Thick, strong and elastic; three layers, with a thick middle layer of smooth muscle and elastic tissue | Thinner wall; the same three layers, but the middle layer is thin and has less muscle and elastic tissue | Only a single layer of endothelium |
| Lumen | Narrow | Broad | Extremely narrow, only one cell wide |
| Valves | Absent | Present in most veins to prevent backflow of blood | Absent |
| Blood carried in adults | Oxygenated, except the pulmonary artery | Deoxygenated, except the pulmonary veins | Changes from oxygenated to deoxygenated as it passes through the tissue |
| Blood pressure | High; blood flows fast | Low; blood flows slowly | Very low and very slow, which allows exchange of materials |

## D. Inquisitive questions

### 1. Why is oxygenated blood separated from deoxygenated blood in heart?

Oxygenated blood contains a high concentration of oxygen and a low concentration of carbon dioxide, while deoxygenated blood is just the opposite. If these two kinds of blood got mixed inside the heart, the blood sent to the body would carry less oxygen and the cells would not get enough oxygen for cellular respiration. The septum, a muscular wall between the left and the right sides of the heart, keeps the two kinds of blood completely separate. Because of this separation, the left side can pump oxygen-rich blood to the whole body at high pressure, while the right side pumps oxygen-poor blood to the lungs at low pressure. In this way every tissue receives fully oxygenated blood.

### 2. How does regular exercise strengthen the circulatory system?

During exercise the muscles need more oxygen, so the heart beats faster and pumps more blood with every beat. When this happens regularly, the heart muscle becomes thicker and stronger, and it can pump more blood with less effort, so the resting heart rate becomes lower. Exercise also keeps the blood vessels elastic, helps to maintain normal blood pressure and cholesterol levels, reduces body fat and improves the circulation to all parts of the body. In this way regular exercise protects a person from cardiovascular diseases such as coronary heart disease, myocardial infarction and high blood pressure.

### 3. How can lifestyle choices prevent cardiovascular diseases?

Cardiovascular diseases are caused mainly by the narrowing and hardening of the arteries due to fatty plaques, high blood pressure, obesity and smoking. These causes are largely related to our daily habits, so they can be prevented by changing the lifestyle. A person should eat a healthy, balanced diet with less fat, sugar and salt and more fruits, vegetables and fibre, should exercise regularly, should not smoke, and should avoid excessive use of alcohol. Keeping the weight normal, controlling blood pressure, cholesterol and diabetes, taking enough sleep and managing stress are also very important. Regular medical check-ups help to detect the problem in the early stages.

### 4. If cholesterol levels in the blood are never regulated, how can it impact the heart and blood vessels over time?

Cholesterol is a fat-like substance which is carried in the blood as low-density lipoprotein (LDL). If its level is never controlled, the excess LDL is deposited on the inner walls of the arteries, and fatty plaques are formed. The arteries become narrow and hard, a condition called atherosclerosis. Because of the narrow lumen, the blood pressure rises and the heart has to work harder, and the damaged walls also make the blood more liable to clot. Over time this causes coronary heart disease, angina (chest pain on exertion), and if a coronary artery is completely blocked, myocardial infarction (heart attack). It may also lead to heart failure and sudden cardiac arrest.

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